

CHAPTER C28

WIND LOADS ON BUILDINGS: MAIN WIND FORCE RESISTING SYSTEM (ENVELOPE PROCEDURE)

The Envelope Procedure is the former “Low-Rise Buildings” provision in Method 2 of ASCE 7-05 for Main Wind Force Resisting System (MWFRS). The simplified method in this chapter is derived from the MWFRS provisions of Method 2 and is intended for simple diaphragm buildings up to 60 ft (18.3 m) in height.

PART 1: ENCLOSED AND PARTIALLY ENCLOSED LOW-RISE BUILDINGS

C28.3 WIND LOADS: MAIN WIND FORCE RESISTING SYSTEM

C28.3.1 Design Wind Pressure for Low-Rise Buildings. See commentary to Section C26.10.1.

C28.3.2 Parapets. See commentary to Section C26.10.2.

Loads on Main Wind Force Resisting Systems: The pressure coefficients for MWFRS are basically separated into two categories:

1. Directional Procedure for buildings of all heights (Fig. 27.3-1) as specified in Chapter 27 for buildings meeting the requirements specified therein; and)]
2. Envelope Procedure for low-rise buildings (Fig. 28.3-1) as specified in Chapter 28 for buildings meeting the requirements specified therein.))]

In generating these coefficients, two distinctly different approaches were used. For the pressure coefficients given in Fig. 27.3-1, the more traditional approach was followed and the pressure coefficients reflect the actual loading on each surface of the building as a function of wind direction, namely, winds perpendicular or parallel to the ridge line.

For low-rise buildings, however, the values of (GC_{pf}) represent “pseudo” loading conditions that, when applied to the building, envelop the desired structural actions (bending moment, shear, thrust) independent of wind direction. To capture all appropriate structural actions, the building must be designed for all wind directions by considering in turn each corner of the building as the windward or reference corner shown in the eight sketches of Fig. 28.3-1. At each corner, two load patterns are applied, one for each wind direction range. The end zone creates the required structural actions in the end frame or bracing. Note also that for all roof slopes, all eight load cases must be considered individually to determine the critical loading for a given structural assemblage or component thereof.

To develop the appropriate pseudovalues of (GC_{pf}), investigators at the University of Western Ontario (Davenport et al. 1978) used an approach that consisted essentially of permitting the

building model to rotate in the wind tunnel through a full 360° while simultaneously monitoring the loading conditions on each of the surfaces (Fig. C28.3-1). Both Exposures B and C were considered. Using influence coefficients for rigid frames, it was possible to spatially average and time average the surface pressures to ascertain the maximum induced external force components to be resisted. More specifically, the following structural actions were evaluated:

1. Total uplift;)]
2. Total horizontal shear;)]
3. Bending moment at knees (two-hinged frame);)]
4. Bending moment at knees (three-hinged frame); and)]
5. Bending moment at ridge (two-hinged frame).)]

The next step involved developing sets of pseudopressure coefficients to generate loading conditions that would envelop the maximum induced force components to be resisted for all possible wind directions and exposures. Note, for example, that the wind azimuth producing the maximum bending moment at the knee would not necessarily produce the maximum total uplift. The maximum induced external force components determined for each of the preceding five categories were used to develop the coefficients. The end result was a set of coefficients that represent fictitious loading conditions but that conservatively envelop the maximum induced force components (bending moment, shear, and thrust) to be resisted, independent of wind direction.

The original set of coefficients was generated for the framing of conventional pre-engineered buildings, that is, single-story, moment-resisting frames in one of the principal directions and bracing in the other principal direction. The approach was later extended to single-story, moment-resisting frames with interior columns (Kavanagh et al. 1983).

Subsequent wind tunnel studies (Isyumov and Case 1995) have shown that the (GC_{pf}) values of Fig. 28.3-1 are also applicable to low-rise buildings with structural systems other than moment-resisting frames. That work examined the instantaneous wind pressures on a low-rise building with a 4:12 pitched gable roof and the resulting wind-induced forces on its MWFRS. Two different MWFRSs were evaluated. One consisted of shear walls and roof trusses at different spacings. The other had moment-resisting frames in one direction, positioned at the same spacings as the roof trusses, and diagonal wind bracing in the other direction. Wind tunnel tests were conducted for both Exposures B and C. The findings of this study showed that the (GC_{pf}) values of Fig. 28.3-1 provided satisfactory estimates of the wind forces for both types of structural systems. This work confirms the validity of Fig. 28.3-1, which reflects the combined action of wind pressures on different external surfaces of a building and thus takes advantage of spatial averaging.

In the original wind tunnel experiments, both B and C exposure terrains were checked. In these early experiments, Exposure

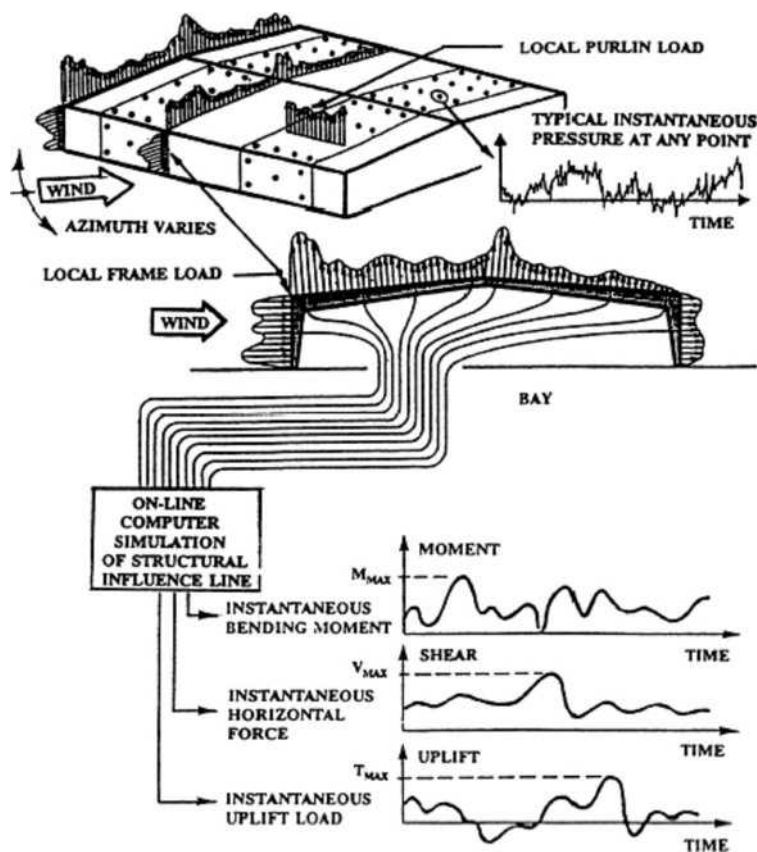


FIGURE C28.3-1 Unsteady Wind Loads on Low Buildings for Given Wind Direction

B did not include nearby buildings. In general, the force components, bending moments, and so forth were found comparable in both exposures, although (GC_{pf}) values associated with Exposure B terrain would be higher than those for Exposure C terrain because of reduced velocity pressure in Exposure B terrain. The (GC_{pf}) values given in Figs. 28.3-1, 30.3-1, 30.3-2A–C, 30.3-3, 30.3-4, 30.3-5A–B, and 30.3-6 are derived from wind tunnel studies modeled with Exposure C terrain. However, they may also be used in other exposures when the velocity pressure representing the appropriate exposure is used.

In comprehensive wind tunnel studies conducted by Ho at the University of Western Ontario (1992), it was determined that when low buildings [$h < 60$ ft ($h < 18.3$ m)] are embedded in suburban terrain (Exposure B, which included nearby buildings), the pressures in most cases are lower than those currently used in existing standards and codes, although the values show a very large scatter because of high turbulence and many variables. The results seem to indicate that some reduction in pressures for buildings located in Exposure B is justified. The Task Committee on Wind Loads believes that it is desirable to design buildings for the exposure conditions consistent with the exposure designations defined in the standard. In the case of low buildings, the effect of the increased intensity of turbulence in rougher terrain (i.e., Exposure B versus C) increases the local pressure coefficients. Beginning in ASCE 7-98, the effect of the increased turbulence intensity on the loads is treated with the truncated profile. Using this approach, the actual building exposure is used and the profile truncation corrects for the underestimate in the loads that would be obtained otherwise.

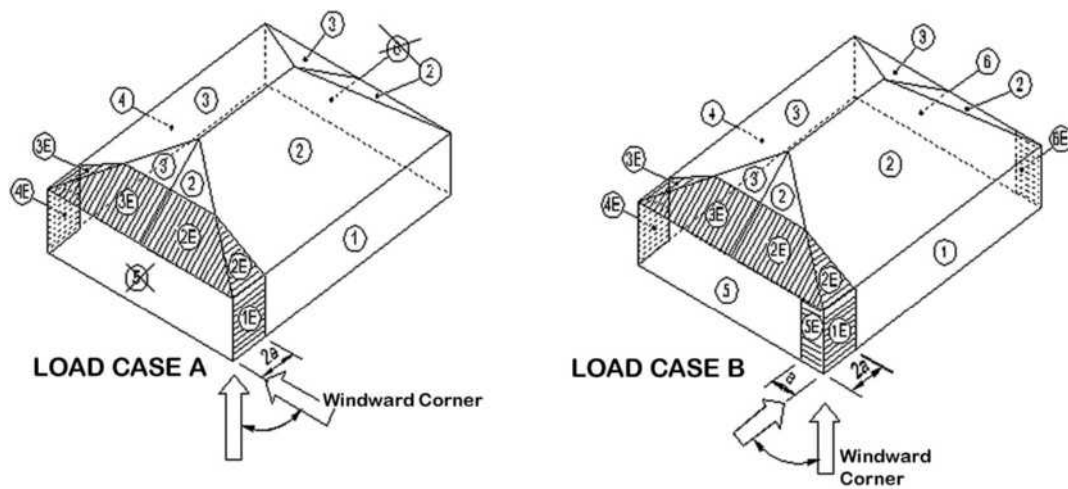
Fig. 28.3-1 is most appropriate for low buildings with width greater than twice their height and a mean roof height that does not exceed 33 ft (10 m). The original database included low buildings with widths no greater than 5 times their eave heights,

and eave height did not exceed 33 ft (10 m). In the absence of more appropriate data, Fig. 28.3-1 may also be used for buildings with mean roof height that does not exceed the least horizontal dimension and is less than or equal to 60 ft (18.3 m). Beyond these extended limits, Fig. 27.3-1 should be used.

All the research used to develop and refine the low-rise building method for MWFRS loads was done on gable-roofed buildings. In the absence of research on hip-roofed buildings, the committee has developed a rational method of applying Fig. 28.3-1 to hip roofs based on its collective experience, intuition, and judgment. This suggested method is presented in Fig. C28.3-2.

Research (Isyumov 1983; Isyumov and Case 2000) indicated that the low-rise method alone underestimates the amount of torsion caused by wind loads. In ASCE 7-02, Note 5 was added to Fig. 28.3-1 to account for this torsional effect and has been carried forward through subsequent editions. The reduction in loading on only 50% of the building results in a torsional load case without an increase in the predicted base shear for the building. This reduction in loading results in equivalent torsion that agrees well with the wind tunnel measurements carried out by Elsharawy et al. (2012, 2015) and Stathopoulos et al. (2013). In general, the provision will have little or no effect on the designs of MWFRS that have well-distributed resistance. However, it will affect the design of systems with centralized resistance, such as a single core in the center of the building. An illustration of the intent of the note on two of the eight load patterns is shown in Fig. 28.3-1. All eight patterns should be modified in this way as a separate set of load conditions in addition to the eight basic patterns.

Internal pressure coefficients (GC_{pi}) to be used for loads on MWFRS are given in Table 26.13-1. The internal pressure load can be critical in one-story, moment-resisting frames and in



Notes

1. Adapt the loadings shown in Fig. 28.3-1 for hip-roofed buildings as shown above. For a given hip roof pitch, use the roof coefficients from the Case A table for both Load Case A and Load Case B.
2. The total horizontal shear shall not be less than that determined by neglecting the wind forces on roof surfaces.

FIGURE C28.3-2. Hip-Roofed Low-Rise Buildings

the top story of a building where the MWFRS consists of moment-resisting frames. Loading cases with positive and negative internal pressures should be considered. The internal pressure load cancels out in the determination of total lateral load and base shear. The designer can use judgment in the use of internal pressure loading for the MWFRS of high-rise buildings.

The edge strip definition was modified following research (Elsharawy et al. 2014) showing that the definition of dimension “a” in ASCE 7-10 led to unduly large edge strips and end zones for very large buildings.

C28.3.4 Minimum Design Wind Loads. This section specifies a minimum wind load to be applied horizontally on the entire vertical projection of the building, as shown in Fig. C27.1-1. This load case is to be applied as a separate load case in addition to the normal load cases specified in other portions of this chapter.

C28.3.5 Horizontal Wind Loads on Open or Partially Enclosed Buildings with Transverse Frames and Pitched Roofs. In 2016, new provisions have been added for wind loads on the longitudinal MWFRS of open or partially enclosed buildings with pitched roofs as shown in Fig. 28.4-3, based on research at the University of Western Ontario (UWO) (Kopp, Galsworthy, and Oh 2010). This study examined the shielding effect of these multiple transverse frames for an open-sided building that was covered by a roof. The shielding effect adopted in ASCE 7 was conservatively simplified; therefore, the exponential form provided in Kopp, Galsworthy, and Oh is a more accurate and acceptable alternative. Building models consisting of three, six, and nine frames were evaluated. A building with two frames was not tested. Therefore, an extrapolation using $n = 2$ is not necessarily conservative. However, this method can be conservatively used for a building with two frames by using $n = 3$. Examples of evaluating these additional wind forces, are presented in Shoemaker, Kopp, and Galsworthy (2011).

The wind loads calculated using Section 28.4.5 are applicable to buildings with open end walls, end walls with the gable filled with cladding, and with additional end wall cladding; however, the area used is always the total end wall area, A_f . The effective solid area of a frame, A_s , is the projected area of any portion of the end wall that would be exposed to the wind.

The measured peak base shear coefficients were used as the basis for the design drag loads in the direction parallel to the ridge

(i.e., wind directions in the range 0° to 45°). These loads include both the effects of friction drag and pressure drag. However, to put this in a format consistent with ASCE 7-10, it was reasonable to use with the enclosed pseudo load coefficients (GC_w) and then apply factors to account for the parameters that affect the load coefficients on open buildings (i.e., building size, solidity ratio, and number of frames. This method yielded conservative results for all experimental wind tunnel data points. The force from Eq. 28.3-3, calibrated to the measured base shear, does not reflect a direct load path from the calculated end wall pressure but is to be used to calculate the longitudinal bracing requirement. For the building configurations evaluated, the UWO study showed that the force measured in the bracing was equal to 70% of the total base shear. The remaining base shear was transferred directly at the column bases.

The wind tunnel studies used to develop the provisions of Section 28.3.5 did not evaluate the effect of obstructed flow due to materials or objects sheltered by the building. Barring an unusual arrangement of materials that could produce a venturi effect, it is judged that obstructed flow would decrease the wind loads on the longitudinal MWFRS. However, as noted in previous studies (Altman 2001, Uematsu and Stathopoulos 2003), the roof wind loads are more sensitive to the effect of obstructed flow.

The wind load in the transverse direction (perpendicular to the ridge) for this type of open building is a separate loading case and is due to the horizontal pressure from the roof load calculated using Section 27.3.2, with C_N from Fig. 27.3-5, and additional pressures acting on the projected areas of any surfaces exposed to the transverse wind.

PART 2: ENCLOSED SIMPLE DIAPHRAGM LOW-RISE BUILDINGS

This simplified approach of the Envelope Procedure is for the relatively common low-rise ($h \leq 60$ ft [$h \leq 18.3$ m]), regular-shaped, simple diaphragm building case (see definitions for “simple diaphragm building” and “regular-shaped building”) where pressures for the roof and walls can be selected directly from a table. Fig. 28.5-1 provides the design pressures for MWFRS for the specified conditions. Values are provided for enclosed buildings only [$(GC_{pi}) = \pm 0.18$].

Horizontal wall pressures are the net sum of the windward and leeward pressures on vertical projection of the wall. *Horizontal*

roof pressures are the net sum of the windward and leeward pressures on vertical projection of the roof. Vertical roof pressures are the net sum of the external and internal pressures on the horizontal projection of the roof.

Note that for the MWFRS in a diaphragm building, the internal pressure cancels for loads on the walls and for the horizontal component of loads on the roof. This is true because when wind forces are transferred by horizontal diaphragms (e.g., floors and roofs) to the vertical elements of the MWFRS (e.g., shear walls, X-bracing, or moment frames), the collection of wind forces from windward and leeward sides of the building occurs in the horizontal diaphragms. Once transferred into the horizontal diaphragms by the vertically spanning wall systems, the wind forces become a net horizontal wind force that is delivered to the lateral-force-resisting elements of the MWFRS. There should be no structural separations in the diaphragms. Additionally, there should be no girts or other horizontal members that transmit significant wind loads directly to vertical frame members of the MWFRS in the direction under consideration. The equal and opposite internal pressures on the walls cancel each other in the horizontal diaphragm. This simplified approach of the Envelope Procedure combines the windward and leeward pressures into a net horizontal wind pressure, with the internal pressures canceled.

The user is cautioned to consider the precise application of windward and leeward wall loads to members of the roof diaphragm where openings may exist and where particular members, such as drag struts, are designed. The design of the roof members of the MWFRS for vertical loads is influenced by internal pressures. The maximum uplift, which is controlled by Load Case B, is produced by a positive internal pressure. At a roof slope of approximately 28° and above, the windward roof pressure becomes positive, and a negative internal pressure used in Load Case 2 in the table may produce a controlling case. From 25° to 45°, both positive and negative internal pressure cases (Load Cases 1 and 2, respectively) must be checked for the roof.

For the designer to use this method for the design of the MWFRS, the building must conform to all of the requirements listed in Section 26.8.2; otherwise, the Directional Procedure, Part 1 of the Envelope Procedure, or the Wind Tunnel Procedure must be used. This method is based on Part 1 of the Envelope Procedure, as shown in Fig. 28.3-1, for a specific group of buildings (simple diaphragm buildings). However, the torsional loading from Fig. 28.3-1 is deemed to be too complicated for a simplified method. The last requirement in Section 28.5.2 prevents the use of this method for buildings with lateral systems that are sensitive to torsional wind loading.

Note 5 of Fig. 28.3-1 identifies several building types that are known to be insensitive to torsion and may therefore be designed using the provisions of Section 28.5. Additionally, buildings whose lateral resistance in each principal direction is provided by two shear walls, braced frames, or moment frames that are spaced apart a distance not less than 75% of the width of the building measured normal to the orthogonal wind direction, and other building types and element arrangements described in Section 27.5.1 or 27.5.2 are also insensitive to torsion. This property could be demonstrated by designing the building using Part 1 of Chapter 28, Fig. 28.3-1, and showing that the torsion load cases defined in Note 5 do not govern the design of any of the lateral resisting elements. Alternatively, it can be demonstrated within the context of Part 2 of Chapter 28 by defining torsion load cases based on the loads in Fig. 28.5-1 and reducing the pressures on one-half of the building by 75%, as described in Fig. 28.3-1, Note 5. If none of the lateral elements are governed by these torsion cases, then the building can be designed using Part 2 of Chapter 28; otherwise, the building must be designed using Part 1 of Chapter 27 or Part 1 of Chapter 28.

Values are tabulated for Exposure B at $h = 30$ ft (9.1 m), and $K_{zt} = 1.0$. Multiplying factors are provided for other exposures and heights. The following values have been used in preparation of Fig. 28.5-1: $h = 30$ ft (9.1 m), Exposure B, $K_z = 0.70$, $K_d = 0.85$, $K_{zt} = 1.0$, $(GC_{pi}) = \pm 0.18$ (enclosed building).

Pressure coefficients are from Fig. 28.3-1.

Wall elements resisting two or more simultaneous wind-induced structural actions (e.g., bending, uplift, or shear) should be designed for the interaction of the wind loads as part of the MWFRS. The horizontal loads in Fig. 28.5-1 are the sum of the windward and leeward pressures and are therefore not applicable as individual wall pressures for the interaction load cases. Design wind pressures, p_s for zones A and C, should be multiplied by +0.85 for use on windward walls and by -0.70 for use on leeward walls (the plus sign signifies pressures acting toward the wall surface). For sidewalls, p_s for zone C multiplied by -0.65 should be used. These wall elements must also be checked for the various separately acting (not simultaneous) components and cladding (C&C) load cases.

Main wind force resisting roof members spanning at least from the eave to the ridge or supporting members spanning at least from the eave to the ridge are not required to be designed for the higher end zone loads. The interior zone loads should be applied because of the enveloped nature of the loads for roof members.

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